

Description and discussion on DCASE 2026 Task 1: Heterogeneous Audio Classification

Panagiota Anastasopoulou, Dmitry Bogdanov, Lonce Wyse, Frederic Font

Music Technology Group, Universitat Pompeu Fabra, Barcelona, Spain

panagiota.anastasopoulou@upf.edu, lonce.wyse@upf.edu, dmitry.bogdanov@upf.edu, frederic.font@upf.edu

Abstract—The Detection and Classification of Acoustic Scenes and Events (DCASE) 2026 Challenge Task 1 *Heterogeneous Audio Classification* is a new task introduced in 2026 which focuses on the automatic classification of heterogeneous audio using the Broad Sound Taxonomy (BST), a general-purpose audio taxonomy that features 5 top-level and 23 second-level categories. The goal of this task is to evaluate sound classification models on diverse, real-world audio that varies widely in its nature, including duration and recording conditions. To that end, two complementary datasets were provided: an expert curated set, BSD10k-v1.2, and a larger crowd-sourced collection, BSD35k-CS, which presents real-world labeling variability. Participants were encouraged to explore audio-based, metadata-based, and multimodal approaches to sound classification, as well as to leverage hierarchical relationships between taxonomy categories. We received a total of 41 submissions from 12 teams. The highest scoring submissions include multimodal systems which take advantage of several pre-trained models for audio and text representation, using techniques for metadata processing and leveraging the two development sets. Overall, the best-scoring submitted system features an increase of $\sim 13\%$ above the provided baseline.

Index Terms—sound classification, hierarchical classification, multi-modal, heterogeneous data, DCASE Challenge

1. INTRODUCTION

Automatic audio classification is a highly relevant task in the field of machine listening and computational audio, which is crucial for many other audio-related tasks and applications such as multimedia organization, creation and retrieval, environmental monitoring, speech recognition, healthcare, security and context-aware computing. Nevertheless, the nature of the automatic classification task can vary widely depending on the audio classes that are used as the target taxonomy, and each specific use case might require different strategies to be successfully accomplished. In particular, the difficulty of a classification task increases when the target classes are highly heterogeneous, requiring classifiers to handle, for example, both short and long sounds, natural and artificial sounds, isolated sounds and mixtures of sound sources, classes ranging from low to high acoustic variability. In such cases, leveraging the class hierarchy relationships derived from the taxonomy and complementing the audio with textual metadata can aid the classification process [1].

This year’s Detection and Classification of Acoustic Scenes and Events (DCASE) Challenge Task 1, *Heterogeneous Audio Classification*¹, focuses on the aforementioned challenges. The remainder of this paper describes the task and its setup, and provides a discussion about the results obtained after analyzing the 41 system submissions received from the 12 teams that participated in the task.

2. TASK DESCRIPTION AND RESEARCH FOCUS

DCASE 2026 Challenge Task 1 aims to evaluate systems for multi-class audio classification across a small but heterogeneous set of sound categories. To that end, the Broad Sound Taxonomy (BST) [2] is used, which contains 5 top-level and 23 second-level categories, and

the goal of the classification task is to correctly predict ground truth labels for the second level of the taxonomy. The BST is designed to represent comprehensively any type of sound at a broad level rather than focusing on detailed sound classes. Unlike other taxonomies which feature more complex hierarchies (e.g. AudioSet [3], Universal Category System [4]), the BST aims to be manageable and user-friendly, accommodating different user profiles such as those in sound design, music production, audio research, and audio practitioners in general.²

To successfully classify audio using the BST, a classifier must be capable of handling audio signals that vary widely in nature, including varying length and a large range of recording and synthesis conditions. As classifiers can benefit from textual metadata in a multimodal setting that complements the acoustic signal, we include further semantic context in the datasets provided for the task (Sec. 3.1). The task also encourages participants to develop classifiers that take advantage of the hierarchical relationships between the sound categories of the taxonomy, and uses an evaluation metric which features bigger penalization for classification errors that miss both levels of the hierarchy over those that only miss the second level (Sec. 3.3).

By addressing a broad sound classification problem using the Broad Sound Taxonomy, this task expects to be more approachable and interpretable than other classification problems based on more complex taxonomies. Task participants can become familiar with all sound categories, facilitating not only the development of models, but also a deeper qualitative analysis and understanding of the results. In particular, this task setup addresses the following scientific questions:

- To what extent can classification models perform successfully across a highly heterogeneous set of audio categories and handle the broad range of variations present in real-world data?
- How can the two-level hierarchical structure of the taxonomy be leveraged to improve classification performance and minimize classification errors that miss both levels of the hierarchy?
- How can noisy, heterogeneous audio data be leveraged to improve model performance and/or generalization?
- How do audio-only models compare with multimodal models for this classification task?

The task of *Heterogeneous Audio Classification* using the BST is potentially useful for diverse practical applications that need to handle a wide range of input sounds by providing an initial, broad, classification stage. As an example of that, an automatic classifier similar to the baseline system provided for this task (Sec. 3.2), has already been deployed in the Freesound³ [5] platform and provides search filtering options based on the BST.

²For a complete list of the BST classes and their hierarchy, we refer the reader to the task description page linked above.

³<https://freesound.org>

¹<https://dcase.community/challenge2026/task-heterogeneous-audio-classification/>

3. TASK SETUP

3.1. Datasets

We leverage Freesound as the data source for the development and evaluation datasets. We provide two complementary datasets for system development annotated with single labels from the Broad Sound Taxonomy. One dataset is smaller and curated by experts, while the other is bigger with crowd-sourced annotations. The evaluation dataset, used for system ranking, also consists of sounds retrieved from Freesound and annotated by experts with BST single labels.

All datasets provide audio files in single-channel, 44.1 kHz, 24-bit format, with maximum duration cropped to 30s. In addition to the audio, the datasets include sound *title*, *tags*, and *description* as textual metadata that can be used in a multimodal setting, and further complementary information such as sound ID, author username, and sound license (which should not be used for training). As a further complement, the datasets also bundle precomputed LAION-CLAP embeddings [6], extracted separately from the audio and the textual metadata, for *quick* experimentation. The text embeddings are extracted using a combination of the sound title, tags, and description as input. Note that development datasets do not include data splits, and participants are encouraged to create their own. For training our baseline system (Sec. 3.2), we follow a 5-fold cross validation approach and report the average of the evaluation metrics across folds. The same strategy can be replicated by participants, including the same random seed, for full reproducibility.

The use of external data and pretrained models is allowed in this task. However, to avoid any overlap with the evaluation dataset, participants should make sure that no Freesound data uploaded after April 1st 2025 is used directly in their training pipeline or indirectly through the use of pretrained models. Note that at the time the challenge was run, none of the widely used Freesound-based datasets and pretrained models (e.g. FSD50k [7], ESC-50 [8], LAION-Audio-630K [6], WavCaps [9], etc.) used Freesound data uploaded after that cutoff date. The following subsections provide more information on each of the datasets provided for this task.

3.1.1. BSD10k-v1.2 (Broad Sound Dataset 10k, v1.2): This dataset was first introduced at the DCASE 2024 Workshop, and consists of a collection of expert annotated sounds aligned with the second level of the BST [10]. For this task, we provide an updated version of the dataset that includes some fixed annotations. The dataset is designed to ensure sufficient diversity within each class. It comprises approximately 11,000 sounds and it is carefully curated to provide well-polished annotations, albeit with an unbalanced distribution across categories. The dataset has a total duration of ~ 35 hours, with individual sounds up to 30 seconds long. In addition to the BST annotations, audio and metadata described above, this dataset provides an annotation confidence score for every sound. More information about the dataset is provided in the corresponding Zenodo page⁴.

3.1.2. BSD35k-CS (Broad Sound Dataset 35k, Crowd-Sourced): The second development dataset contains approximately 35,000 sounds (~ 150 hours of audio) uploaded to Freesound between April 1st, 2025, and January 27th, 2026. Since April 2025, users uploading sounds to Freesound are required to choose an appropriate BST category for each sound, and we use these as ground truth labels for this dataset. Even though the taxonomy is designed to be easy to use, we have observed that variation exists in the way users interpret the taxonomy, and this results in potentially inconsistent annotations and a noisy ground truth. Furthermore, unlike BSD10k, this dataset does

not include annotation confidence scores. More information about the dataset is provided in the corresponding Zenodo page⁵.

3.1.3. DCASE 2026 Task 1 Evaluation set: The evaluation dataset consists of 1288 sounds⁶, expert labeled with the corresponding BST category and uploaded to Freesound between April 1st 2025 and June 1st, 2026. This set contains heterogeneous sounds, generally balanced across the 23 target second-level BST categories. When possible, diversity within each category is also assessed. The dataset does not overlap with the other development datasets. Also, unlike BSD10k, this dataset does not include annotation confidence scores. Note that when the evaluation dataset was originally made available during the challenge, no ground truth labels were published, and only the audio files and descriptive textual metadata (which were needed for running inference with multimodal models) were available. Also note that the original evaluation set also contained *additional* sounds which were not used for computing the evaluation scores, reaching a total of 3246 sounds. More information about the evaluation set is provided in the corresponding Zenodo page⁷.

3.2. Baseline system

As a baseline system, we use variations of the HATR model presented at the DCASE Workshop 2025 [1]. We include a multimodal version and an audio-only version, both *non-hierarchical*⁸ models trained on audio and text representation vectors extracted using the pretrained LAION-CLAP model. Both variants train a small classifier on top of the CLAP representation(s). The multimodal variant complements the audio embeddings with text embeddings extracted from textual sound metadata (title, tags, description), and fuses the two through an attention mechanism. Because the encoders stay frozen, the trainable model is compact ($\sim 3.7M$ and $\sim 7.3M$ parameters) and easy to reproduce. The baseline is available in a public source code repository⁹.

3.3. Evaluation metrics

The task evaluation metrics consist of variations of the standard precision, recall and F-score (P , R , F) which take into account the class hierarchy (hP , hR , and hF , respectively) [11]. These metrics are designed to assign a stronger penalty to misclassifications that are incorrect at both levels of the taxonomy (i.e. confusing two classes with a common parent results in better score than confusing two classes with different parents). We customize the hP , hR , and hF metrics as defined in [11] and add a λ parameter ($\lambda = 0.75$ for this task) to parametrically adjust the score assigned to classifications that are only correct at the top level of the taxonomy (i.e. classifications that are partially correct). A full implementation of the evaluation metrics is provided in the code for the baseline system referenced above. The metric used for the task leaderboard is macro-averaged hF . To give further intuition of how the metric behaves, Table 1 reports hP , hR , and hF scores for a number of simulated systems with different characteristics.

4. TASK RESULTS

Table 2 shows the teams leaderboard for the DCASE Challenge 2026 Task 1. The best submission (Primus_CPIJKU, $hF = 0.836$) improves on the multimodal baseline trained on both development

⁵<https://doi.org/10.5281/zenodo.19187100>

⁶Task participants had to provide predictions for a larger number of sounds, but only 1288 of them were used for evaluation purposes.

⁷<https://doi.org/10.5281/zenodo.20442927>

⁸The hierarchical component of HATR is omitted.

⁹https://github.com/MTG/dcase2026_task1_baseline

⁴<https://doi.org/10.5281/zenodo.17233904>

Table 1: Hierarchical precision (hP), recall (hR), and F-score (hF) for different simulated systems ($\lambda = 0.75$). **Perfect:** a system with perfect prediction capabilities. **Half wrong:** a system wrong half of the time; its errors miss the second level but may still be correct at the top level. **Half wrong (top level correct):** a system wrong half of the time; its errors miss the second level but are always correct at the top level. **Half wrong (both levels wrong):** a system wrong half of the time; its errors miss both levels. **All wrong:** a system always wrong; its errors miss the second level but may still be correct at the top level. **All wrong (top level correct):** a system always wrong; its errors miss the second level but are always correct at the top level. **All wrong (both levels wrong):** a system always wrong; its errors miss both levels. **Random:** a system predicting at random, correct $1/N$ of the time (with N the number of second-level categories); wrong predictions may still be correct at the top level.

Simulated system name	hP	hR	hF
Perfect	1.000	1.000	1.000
Half wrong (top level correct)	0.688	0.688	0.688
Half wrong	0.535	0.534	0.534
Half wrong (both levels wrong)	0.509	0.503	0.504
All wrong (top level correct)	0.375	0.375	0.375
Random	0.118	0.117	0.116
All wrong	0.073	0.073	0.071
All wrong (both levels wrong)	0.000	0.000	0.000

Table 2: DCASE Challenge 2026 Task 1 teams leaderboard.

Rank	Submission code	hP	hR	hF
1	Primus_CPJKU_task1_3	0.842	0.846	0.836
2	Huang_WHU_task1_2	0.825	0.810	0.811
3	Guan_HEU_task1_3	0.816	0.797	0.799
4	Zheng_SCUT_task1_1	0.817	0.782	0.788
5	Kucukoglu_NYU_task1_1	0.824	0.775	0.787
6	Lin_JKU_task1_1	0.801	0.785	0.785
7	Baseline_UPF_task1_2	0.725	0.696	0.700
8	Colotti_TAU_task1_1	0.746	0.709	0.699
9	Kil_Medisensing_task1_1	0.682	0.732	0.692
10	Liu_CQUPT_task1_3	0.657	0.647	0.629
11	Sharma_IR_task1_1	0.526	0.488	0.472
12	Han_CSU_task1_1	0.422	0.370	0.372
13	Zhang_XJTLU_task1_3	0.162	0.139	0.116

datasets ($hF = 0.70$) by 0.136 hF , and six of the twelve teams’ best submissions scored above the baseline. The submissions spanned a broad design space: the winner (Primus_CPJKU) fused an LLM-derived class-probability prior with a multi-backbone ensemble, Guan_HEU fine-tuned Qwen audio-language models with hyperbolic hierarchical embeddings, Huang_WHU combined CLAP with hand-crafted spectral branches and nearest-neighbour retrieval post-processing, and Kil_Medisensing built decision-level posterior stacks with conservative metadata gating. Comparing the leaderboard against the simulated reference systems of Table 1 shows that the top three teams sit well above the “half wrong (top level correct)” reference (0.688), whereas the two lowest submissions fall onto degenerate points; Han_CSU (0.372) near “all wrong (top level correct)” (0.375) and Zhang_XJTLU (0.116) at the random system (0.116). The nature of these errors differs: Zhang_XJTLU concentrates most of its predictions on a handful of high-frequency classes (e.g. *Objects / House appliances*), an almost degenerate output that lands at the random reference, whereas Han_CSU errs at both taxonomy levels rather than only at the leaf, so its proximity to the “top level correct” point is partly coincidental. Together with unusually large gaps between their reported development and evaluation scores, both patterns are more consistent with a submission-format or pipeline fault than with trained model behaviour, and are flagged for further verification.

Almost all teams (except Guan_HEU) built on the frozen LAION-CLAP audio–text model as their representation backbone. The strongest systems added complementary pretrained audio transformers on top (Primus_CPJKU combined PaSST, BEATs, and M2D; Liu_CQUPT added PaSST). Alternatively, Huang_WHU complemented CLAP with hand-crafted log-STFT, log-mel, and MFCC branches, while Guan_HEU departed from CLAP altogether by fine-tuning Qwen audio-language models. Some systems tried replacing the CLAP audio encoder with other alternatives (e.g. Kucukoglu_NYU tried BEATs, ConvNeXt, MATPAC, and Whisper), but these do not seem to yield performance benefits.

Multimodal systems clearly dominated: every submission in the upper half of the leaderboard combined the textual metadata with the audio signal, and all audio-only submissions ranked in the lower half. The best audio-only submission (Liu_CQUPT, $hF = 0.629$) is in fact a multimodal-trained system whose submitted variant drops the text branch at inference, and within that same team the purely audio-only system fell well below its metadata-gated counterparts. The leading teams went beyond encoding the raw metadata with the CLAP text encoder: the winning system (Primus_CPJKU) used a large language model not only to summarize the metadata, but also to build a structured class-probability prior over the taxonomy that was fused with the acoustic representation. Furthermore, Kil_Medisensing gated the metadata against the audio-predicted parent, and Liu_CQUPT trained a continued-pretrained text branch. The metadata was also treated selectively rather than uniformly: Colotti_TAU deliberately used only cleaned tags and descriptions and discarded the sound titles, while Zhang_XJTLU and Liu_CQUPT trained with metadata- and text-dropout variants to avoid over-relying on it. Part of this multimodal advantage is, however, entangled with data provenance (see discussion below).

Regarding hierarchy, teams incorporated the two-level taxonomy in three distinct places: in the training loss (Kucukoglu_NYU, Liu_CQUPT, Sharma_IR, Han_CSU, through auxiliary top-level terms, hierarchical label smoothing, or proxy losses), in the representation (Guan_HEU and Colotti_TAU, through hyperbolic embeddings of the parent–child structure), and at decoding (Kil_Medisensing and Zhang_XJTLU, through parent-guided or “parent-safe” rules that keep the leaf prediction inside the audio-predicted parent). Several of the strongest systems –the winner (Primus_CPJKU), the baseline reproduction (Zheng_SCUT), and Lin_JKU– instead used a flat second-level head with no explicit hierarchical component. The reproduction is itself worth a closer look: it scores well above the official baseline (0.788 versus 0.700) while reporting the same architecture and training only with BSD10k-v1.2 (on which the baseline trained the same way reaches only $hF = 0.59$) and whether this stems from differences in fold ensembling and model selection or from an inadvertent development/evaluation overlap should be verified. The hierarchy is, of course, present throughout, in the evaluation metric and in how most teams decode, so the finding may be narrower than it first appears: adding an explicit hierarchical *training* objective on top of already-strong multimodal feature representation seems to bring little further accuracy gain. This is consistent with the earlier hierarchical-modeling study by the authors [1]. Where hierarchical training did help was in systems based on simpler representations — Kucukoglu_NYU reports that a hierarchical loss improved its audio-only model though not its multimodal one — and, most clearly, at *decoding* time by limiting the leaf category decision within does that belong to the audio-predicted parent (Kil_Medisensing, Zhang_XJTLU).

Task participants used different strategies for leveraging the crowd-sourced BSD35k-CS set, but many decided to process the provided

ground-truth labels instead of using them raw. The winning team trained a teacher model on the curated BSD10k-v1.2 set, relabeled every BSD35k-CS item with the teacher’s soft pseudo-labels, and blended them with the crowd labels at a weight of 0.75–1.0 — effectively overriding the crowd labels, which it found less reliable than the pseudo-labels. Others submissions opted to validate the labels rather than re-estimating them. Huang_WHU teacher-filtered a trusted subset with per-uploader caps, Kil_Medisensing and Zhang_XJTLU reliability-weighted the noisy labels, and Guan_HEU used BSD35k-CS only for coarse parent-level pretraining. Teams that skipped BSD35k-CS (Sharma_IR and Colotti_TAU; trained on BSD10k-v1.2 only) or used its raw labels (Liu_CQUPT), generally scored lower. In the case of the task baseline, adding BSD35k-CS helped improving hF from 0.59 to 0.70, a gain of roughly 0.11. This gain should be read with care: because BSD35k-CS and the evaluation set were collected in the same recent period and share a fraction of Freesound uploaders, part of the benefit may reflect familiarity with those users rather than better generalization (see Sec. 3.1 and also see below). Interestingly enough, Colotti_TAU built an uploader-disjoint partition and showed that a random 5-fold split inflates the development score by roughly twice as much as an uploader-aware one; direct evidence that per-uploader leakage¹⁰, not only label noise, shapes these numbers, so systems trained on BSD10k-v1.2 alone may be under-ranked. More generally, because no fixed development splits were provided, reported development scores were often well above evaluation scores, and the teams whose two scores agreed most closely (notably Primus_CPJKU and Lin_JKU) were those that validated on uploader-disjoint splits and used conservative model selection. Notably, importing *additional* external audio corpora did not seem to help: the three teams that did so (Kil_Medisensing, Liu_CQUPT, Zhang_XJTLU) all ranked in the lower half, two further teams (Kucukoglu_NYU, Lin_JKU) tested external sets and dropped them, and the top three teams used only the two provided development sets. This may stem from the difficulty of mapping external label spaces onto the 23 BST classes rather than from the amount of data, and, more broadly, we cannot know exactly how each team mapped or filtered its data nor how much residual label noise this introduced.

Performance was strongly class-dependent: the instrument-sample leaves are near-saturated (per-class $hF > 0.95$ for most top systems), whereas the hardest classes (e.g. processed/synthetic speech, electronic/design sound effects) remain around 0.60–0.65 even for the winner. Lin_JKU found positive correlation between per-class scores and BSD10k-v1.2 annotation-confidence scores. That suggests that residual errors concentrate on the sounds that human annotators themselves label least confidently, so the score ceiling on these classes is partly intrinsic to label ambiguity. Beyond using confidence as a diagnostic, a few teams fed label reliability back into training — reliability-weighting the crowd labels (Kil_Medisensing, Zhang_XJTLU) or pruning examples by influence (Han_CSU) — although none of these reached the top of the leaderboard.

Nearly every submitted system was an ensemble, but ensemble *diversity* mattered more than ensemble size: the strongest systems (led by Primus_CPJKU) combined structurally different backbones, whereas averaging seeds, checkpoints, or cross-validation folds of a single architecture mainly reduced variance. This distinction matters when reading the scores, since a 5-fold cross-validation ensemble (as in the Zheng_SCUT baseline reproduction) mainly reduces variance across folds of one architecture rather than combining independent

models. Guan_HEU further shows that a single strong model remained competitive — its best submission is a single audio-language model rather than its own ensemble — while Huang_WHU and Lin_JKU report that their ensembles added little over their best single model.

According to the task results, model complexity does not seem to follow performance, and in some cases it works against it. Sharma_IR extends the baseline’s frozen-CLAP late-fusion recipe with a third backbone, a transformer fusion module, a hierarchical loss, and checkpoint ensembling, yet — even aside from its skipping BSD35k-CS — performs below the baseline. Additionally, the two most elaborate pipelines, a 25-model out-of-fold selection (Han_CSU) and a 31-family ensemble with a learned reranker (Zhang_XJTLU), suffered the largest development-to-evaluation performance drops. A likely reason is the nature of the task: with most of the discriminative signal already in the frozen CLAP representation and an evaluation set that differs in distribution from the clean development data, extra trainable or model-selection complexity mainly fits development-specific structure which is not transferable to the evaluation set.

To finish this discussion, it is worth mentioning that several distinctive design choices cannot be credited with confidence, since teams differ along many axes at once (backbones, data curation, hierarchy, ensembling). For instance, the winner’s LLM-derived class prior is possibly its edge, but it is only briefly described. The hyperbolic embeddings of Guan_HEU and Colotti_TAU are appealing for a nested taxonomy, yet with only two levels their benefit is unclear. Retrieval-based post-processing (Huang_WHU) and explicit hierarchical losses seem to be less relevant when a strong audio-text representation is present. We report these findings as unverified, yet promising, research directions.

Complementary information about the task results, including extra figures, system data and the technical reports can be found on the task results page¹¹.

5. CONCLUSION

This paper summarises DCASE Challenge 2026 Task 1, including the task setup and a discussion about the systems submitted and the results obtained. On the questions this edition of the task has addressed: multimodal systems that exploit the textual metadata clearly beat audio-only ones, and the leaders processed that metadata rather than merely encoding it (though audio-only models stay relevant for uses such as annotation assistance); the two-level hierarchy helped more as a decoding constraint than as a training objective; and the noisy crowd-sourced set paid off mainly when its labels were re-estimated or filtered, the clearest differentiator among the leaders. Beyond the rankings, evaluation setup also matters: because the evaluation set draws heavily on BSD35k-CS contributors, future editions would benefit from uploader-disjoint splits and reporting against the reference systems. Overall, with such heterogeneous data, careful data-centric selection is an important complement to model design, and future systems could explore an “other” class under each top-level category to better cover real-world BST applications.

6. ACKNOWLEDGMENT

This research was partially funded by the Secretaria d’Universitats i Recerca del Departament de Recerca i Universitats de la Generalitat de Catalunya (2024FI-100252) under the Joan Oró program, the Càtedra IA y Música project (TSI-100929-2023-1), funded by the Secretaría de Estado de Digitalización e Inteligencia Artificial and European Union-Next Generation EU funds and by BMAT Music Innovators, and

¹⁰The evaluation set shares no individual sounds with the development sets. Same-uploader clips may share style and wording.

¹¹<https://dcase.community/challenge2026/task-heterogeneous-audio-classification-results>

the IMPA project (PID2023-152250OB-I00) funded by the Ministry of Science, Innovation and Universities of the Spanish Government, the Agencia Estatal de Investigación (AEI) and co-financed by the European Union. We would like to thank all the teams who participated in the task.

REFERENCES

- [1] P. Anastasopoulou, F. A. Dal Rí, X. Serra, and F. Font, “Hierarchical and multimodal learning for heterogeneous sound classification,” in *Proc. Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2025.
- [2] P. Anastasopoulou, X. Serra, and F. Font, “A general-purpose sound taxonomy for the classification of heterogeneous sound collections,” In press. [Online]. Available: <https://www.researchsquare.com/article/rs-7206795/v1>
- [3] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.
- [4] T. Nielsen, J. Drury, and K. Paquin, “Universal category system,” <https://universalcategorysystem.com>, accessed: 2026-07-01.
- [5] F. Font, G. Roma, and X. Serra, “Freesound technical demo,” in *Proceedings of the 21st ACM international conference on Multimedia*, 2013, pp. 411–412.
- [6] Y. Wu*, K. Chen*, T. Zhang*, Y. Hui*, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP*, 2023.
- [7] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, “FSD50K: an open dataset of human-labeled sound events,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2022.
- [8] K. J. Piczak, “ESC: Dataset for Environmental Sound Classification,” in *Proceedings of the 23rd Annual ACM Conference on Multimedia*. ACM Press, 2015, pp. 1015–1018. [Online]. Available: <http://dl.acm.org/citation.cfm?doid=2733373.2806390>
- [9] X. Mei, C. Meng, H. Liu, Q. Kong, T. Ko, C. Zhao, M. D. Plumbley, Y. Zou, and W. Wang, “Wavcaps: A chatgpt-assisted weakly-labelled audio captioning dataset for audio-language multimodal research,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, p. 3339–3354, 2024. [Online]. Available: <http://dx.doi.org/10.1109/TASLP.2024.3419446>
- [10] P. Anastasopoulou, J. Torrey, X. Serra, and F. Font, “Heterogeneous sound classification with the Broad Sound Taxonomy and Dataset,” in *Proc. Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*, 2024.
- [11] S. Kiritchenko, S. Matwin, and A. F. Famili, “Functional annotation of genes using hierarchical text categorization,” in *Proc. of the ACL Workshop on Linking Biological Literature, Ontologies and Databases: Mining Biological Semantics*, 2005.